



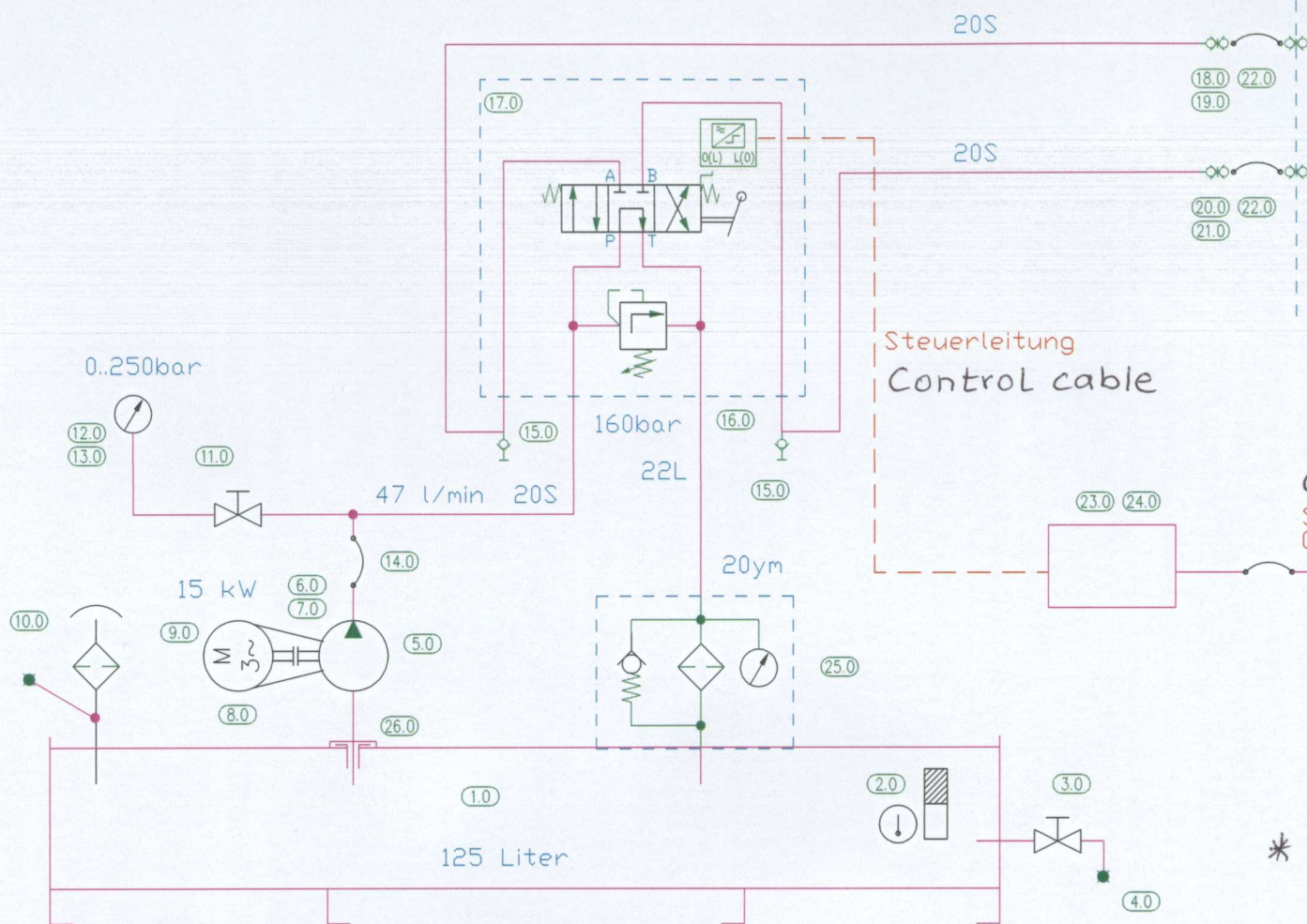
Deckblatt Auftragsdokumentation
Coverage Documentation

Gegenstand	:	Pumpenkuli HSAV 125/50
OBJECT		MOBILE PUMP TROLLEY
Loesche Bestell-Nr.	:	4501.8612
LOESCHE ORDER-NO.		
Kennwort	:	Narayangonj
KEYWORD		
Hersteller Auftrags-Nr.	:	107237/1
MANUFACTURE ORDER-NO.		
Erstellungsdatum	:	2004
MANUFACTURING DATE		

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ADJUST HANDVALVE

Ersatzteilliste			RUHFUS Systemhydraulik 
Spare Parts List			
Pumpenkuli HSAV125/50 80.1250.1/0			Kennwort : Hereke 5 RM
Mobil pump trolley			Keyword
Pos	Stk	Benennung	Artikel-Nr.
Item	qty.	Description	Part-No.
1	1	Ölbehälter kpl.	A58öIHSAV12550
		Oil tank cpl.	
2	1	Flüssigkeitsanzeige	A05700076
		Liquid indicator with thermometer	
3	1	2/2 Wege-Kugelhahn PN16 1/2"	A32700204
		2/2-direction ball valve	
4	1	Verschlußschraube	A7228976
		plug	
5	1	Zahnradpumpe	A61PLP30*34
		Gear pump external	
6	1	Kupplung	A03A384542H7N3Alu
		Coupling	
7	1	Pumpenträger	A03PT3500216
		Pump holder	
8	2	Dämpfungsschiene	A03DL0007
		Damping ring	
9	1	Elektromotor 4-polig	A15CC16L4200003
		Electro motor	
10	1	TankbelüftungsfILTER mit Einfüllsieb	A05320037
		Tank breather with input filler	
11	1	Manometerabsperrentil	A34100041329
		Pressure gauge isolator valve	
12	1	Manometer mit Glyzerin	C58GMU630250
		Pressure gauge filled with glycerin	
13	1	Manometer Anschluß Verschraubung	A9300262049
		Pressure gauge fitting	
14	1	Schlauch 2SN12	A742SN16230DKOS20045
		Flexible hose	
15	2	T-Messkupplung	A762103122000
		Straight coupling with plug-in coupling	
16	4	Gerade Einschraubverschraubung	A93GE16SREDA3C
		Male stud coupling	
17	1	Handhebelventil mit Mikroschalter	A61VCD50118MS
		Hand lever valve with micro switch	
18	2	Schnellkupplung HD-Stecker	C50KS11316003LPP
		Push pull coupling -quick action	
19	2	Staubschutz Stecker	C50646390310067
		Dust shield plug	
20	4	Schnellkupplung HD-Muffe	C50KM11316003LPP
		Push pull coupling -quick action	
21	4	Staubschutz HD-Muffe	C50646390310066K
		Dust shield	
22	2	Schlauch 2SN16	A742SN165mDKOS200
		Flexible hose	



Aggregatsgrenze
Battery limit

Steuerleitung
Control cable

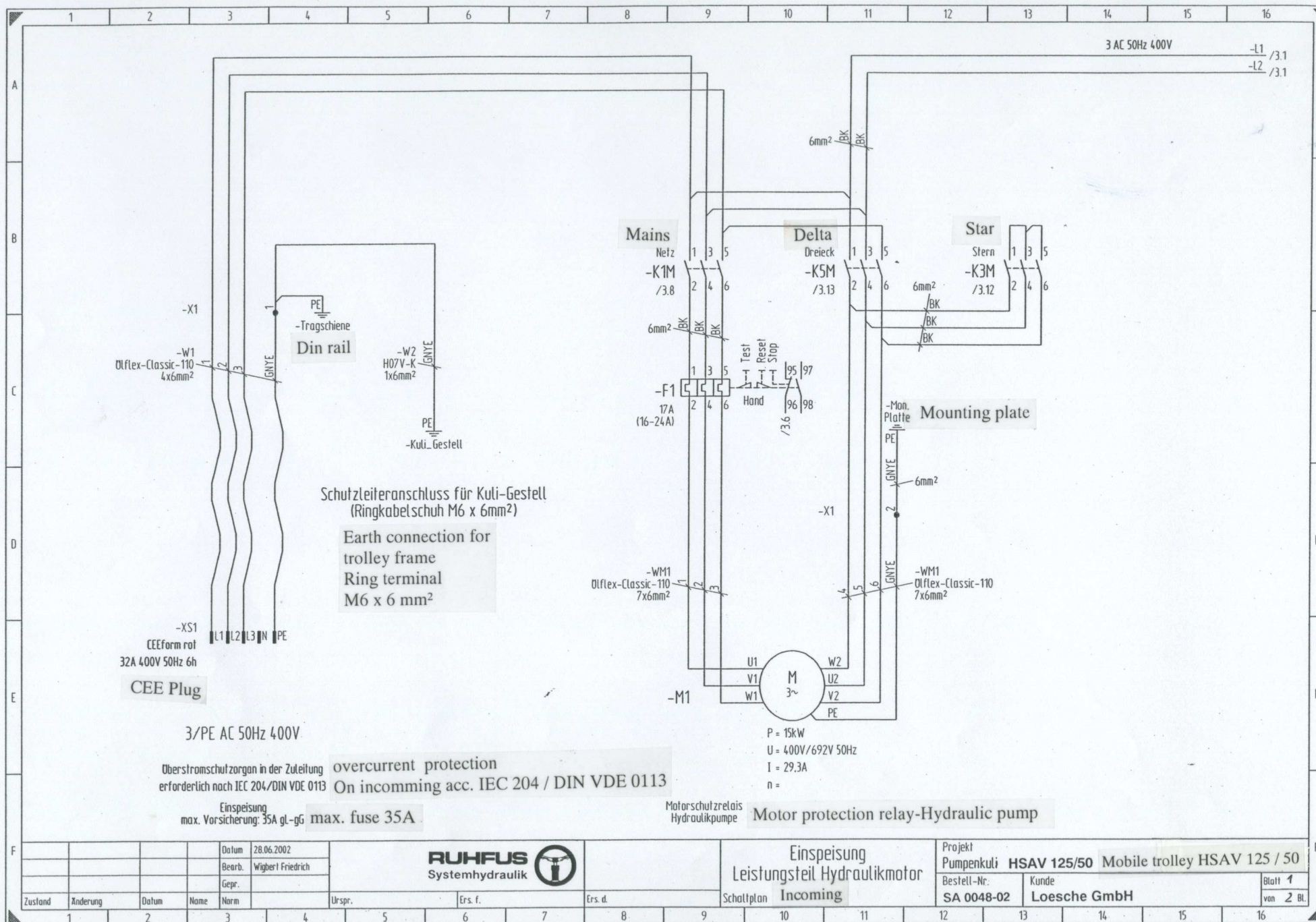
CEE-Plug *
Stecker
CEE 32 Ampere

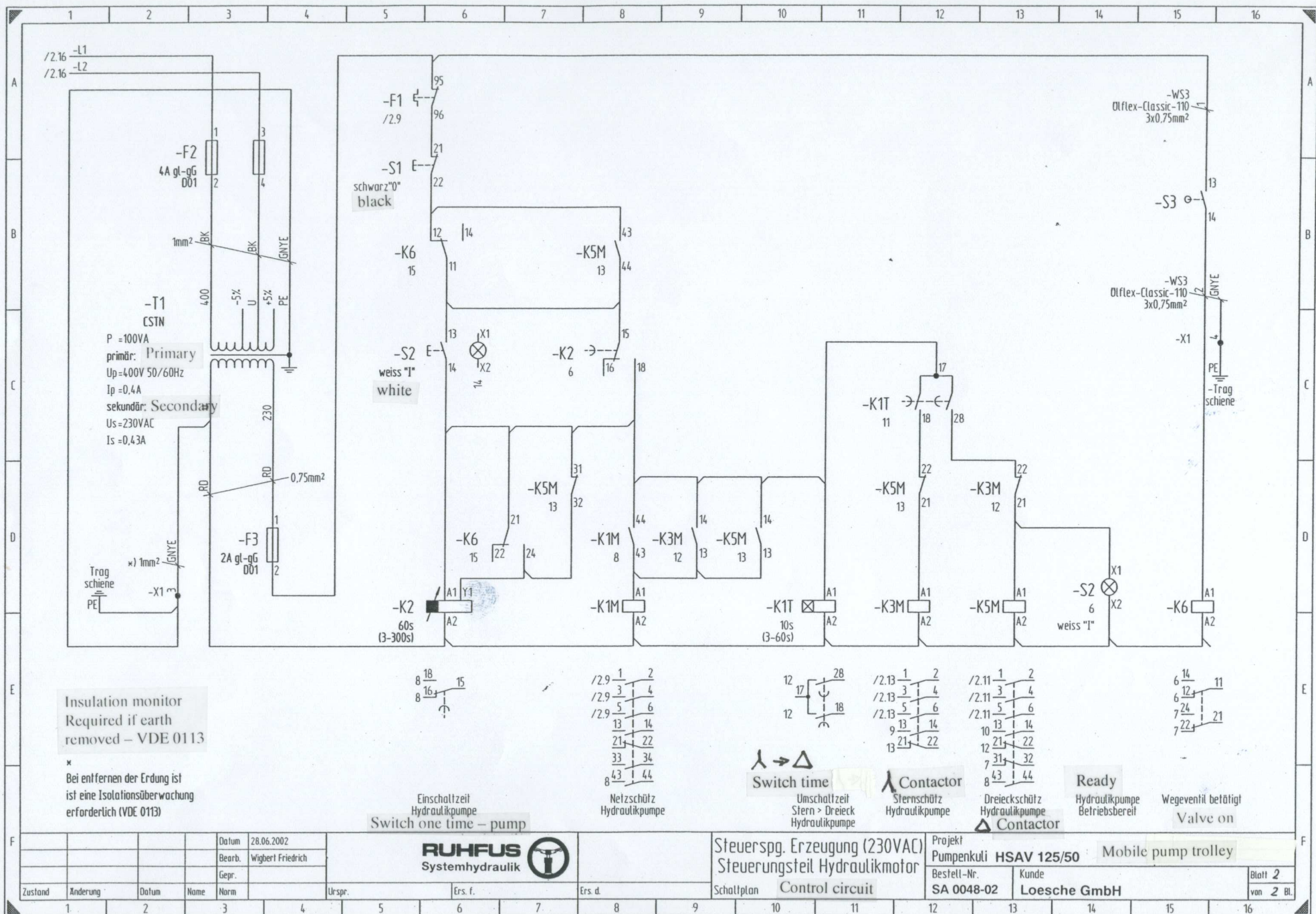
* CEE-Plug may be
replaced by a 3-phase
plug acc. Local standards.

DESIGN AUTHORITY				USED ON / verwendet in					
						Projekt:			
						SA0048			
Benennung: <i>Mobile trolley HSAV</i> Pumpenkuli HSAV 125/50									
SCALE: Maßst:					DATE / Datum NAME / Name	DR. 24.05.02 gez. PLH	CK gepr		SEEN ges
						DRAWING NO: Zeichnungs-Nr:	SA - 0048	- 01	

Material / material: hergestellt aus / manufactured by: ähnliches Teil / similar part:

SCALE:
Maßst:





Transport Regulations

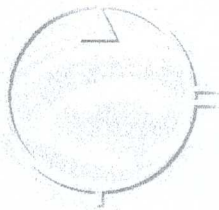
for Hydraulic Units

Prior to transport, the oil system of the respective unit must be depressurised !

Hydraulic accumulators must be emptied !

When lifting the unit, the provided lifting eyes are to be used. Within this context, please observe that the lifting gear does not damage any hydraulic or electrical components.

If a fork lift is used for transport purposes, the forks should be positioned as close to the U profiles beneath the oil tray as possible to prevent the tray from bending.



General Operating- and Maintenance Instructions for Hydraulic Systems

I. Introduction

Our hydraulic system is expected to have a most extensive troublefree operation and a long working life. Therefore we recommend to exactly obey our instructions as follows, i.e. concerning assembly and putting into operation of the system in order to be protected against unnecessary machine down-times and expenses. This instruction is generally valid for Hydraulic power units independent of the respective hydraulic control. Attached please find the service records concerning the components contained in the circuit diagram. The local operating conditions given in the location which has been chosen by you which might influence working time and function of the hydraulic system have to be obeyed separately.

Not only as during installation of the system but also during commissioning and concerning the following inspections in the first place there must be attached importance to extraordinary cleanliness during all working procedures.

Nevertheless we recommend to convince oneself concerning the condition of the aggregate, i.e. immediately after receipt of system, as a contamination or damaging might be possible as by means of improper transportation. As far as an inadmissible contamination can be discovered we kindly ask for immediate information, i.e. before the installation which has to be done. As far as this point will not be obeyed, we cannot bear possible expenses for cleaning and resulted by means of contamination of the individual devices, as the reason of contamination cannot be detected after the installation.

II. Assembly

It has to be obeyed during assembly that pipe outlets, charging holes etc. at the aggregate will remain closed until the final installation of the connecting conduits respectively until filling of the reservoir. For this purpose principally plugs and/or caps made from plastics have to be used, but never any improvised selfmade stopples from textiles or paper.

III. Imbedding of Pipelines (Piping)

As material we recommend seamless precision steel pipes according to DIN 2391 with quality standards (bright annealed, bondered rust-protective and oiled) according to DIN 2391C. As material standard St 35.29 is used. The determination of the nominal widths and wall thicknesses will generally take place during planning process of the total system, i.e. in cooperation with our construction dept. and respective data can be furtherly taken from our circuit diagrams.

As far as no values have been fixed here, the following areas for oil speeds in the pressure line have to be considered concerning the choice of nominal widths.

Concerning the suction channels 0,5 m/s must not be exceeded, and concerning the reflow lines 2,0 m/s and according the following table for the pressure line must not be exceeded.

P [bar]	25	50	100	up to 200	more than 200
V [m/s]	2,5...3	3,5...4	4,5...5	5...6	6...8

The necessary pipe wall thickness has to be fixed according to DIN2413

“Calculation of wall thickness of steel pipes stressed by internal pressure“.

During production make of the pipelines any heat treatment should be comprehensively avoided. Consequently the pipes have to be cold-bended - if possible.

In case that heat treatment cannot be avoided so the welding seams possibly have to be laid so that the internal seams could be smoothend by means of a grinding device. Furthermore the pipes have to be brushed by means of a suitable flexible cleaning device in order to remove remaining welding beads and forging residues. It might be appropriate to stain the prepared pipes with a solution of approx. 10% hydrochloric acid 90% water at approx. 50 °C, i.e. for a period of approx. 2 hours after this, then to rinse severally by means of clear water and to have them dried by means of hot air after this. The prepared pipelines have to be closed up to their final installation by means of plastic stoppers. During installation the pipelines have to be fixed that way, that no inadmissible vibrations can occur. Concerning longer lines pipe clamps have to be used. Within the pipeline network no tensions (tensible forces) must occur.

IV. Embedding of Hose Pipes

For the dimensioning of hose pipes the standards which are mentioned unter III. are valid concerning the nominal width. It is appropriate to use checked hoses which have been (al)ready assembled by the producer.

Concerning quality and type recommendation of the relevant supplier firms as the demands on high pressure hoses are extraordinary different coming up to the respective working conditions. At installation of the flexible lines the instructions of the producer according to the bending radii etc. have to be considered. Folding-, squeezing- and scrubbing spots must be avoided.

V. Type of Oil Reservoir

In case that the oil reservoir is not part of our delivery so we recommend to have the internal space of the reservoir lacquered with an oil-proof paint. Before lacquering the internal walls have to be carefully cleaned from welding forge, rolling skin and rust. We are pleased to give respective information concerning kind and quality of paint.

VI. Filling of the Oil Reservoir

The filling of the reservoir must be done with hydraulic oil and according to our recommendations. The filling must be done via the filling neck which is provided for this only. Before filling-in of fluid it must be obeyed, that the used tanks are clean. Besides the removal of contaminations at the bungs of the storage tanks the liquid must be checked for contamination and for possible water which might have penetrated. This can be done by means of a drop test. After finished filling procedure the filling neck must be closed.

VII. Reservoir Filling

The filling of possibly existing air-hydraulic accumulators has exclusively to be done according to the instructions of the producer. The necessary instructions for this are attached. Before delivery of the systems the storages will be provided in our place with a prefilling pressure amounting to 5...10 kp/cm².

VIII. Preparations for Initial Operation

After finishing of assembly and oil filling of the system the following points have to be checked and carried out before the system will be switched-on:

1. Control of oil level
2. Control of supply- and control voltage
3. Comparison of direction of rotation of pump and motor
4. Opening of relief valves, throttle valves and volume governors
5. Bring possibly provided valves or ball valves in the suction channel into working position.
6. As far as the system is containing pressure storage/s such storage/s and safety equipments have to undergo an inspection test by the TÜV, i.e. before the first commissioning in the using place and according to the rules for prevention of accidents "pressure tanks".

IX. Initial Operation

Following to the points mentioned unter VIII it can be started with the test run. At first the pump shaft has to be turned manually in order to control the running of the pump. Then a screwed connection at the pressure side of the pump will be loosened by means of some rotations and the motor is severally switched-on shortterm in order to find out if the pump will faultlessly suck-on. This tipping operation is especially important concerning wing cell pumps. Concerning axial piston pumps with leakage oil connection the pump must be filled with hydraulic oil before switching-on of the E-motor. After the suction conditions will have been checked the complete system will be filled at minor working pressure and the total system must be ventilated. This can be done by means of slight opening of the ventilating cocks and the ventilating screws which are provided for that. In case that concerning greater systems it will be ventilated in several places so the procedure will be started at the place which is next to the pump and in the most distant ventilating possibility, and will be finished i.e.: at the descending force groups. In general trunk piston cylinders are provided with a ventilating screw. As this is mostly not the case concerning double-acting cylinders the pistons are run severally from end position to end position. The ventilating openings are to be closed when bubble-free oil is flowing out only.

Then the operations of the working devices will be checked individually, i.e. also at a low working pressure. In case that the checking will turn out to be satisfying the system will be slowly brought to working temperature by means of regulation at the pressure control valve.

During the procedure the following has to be permanently obeyed:

1. Control for leakages in the line system
2. Observation of the oil level within the reservoir
3. Observation of working temperature

In case that leakages in the system will be found, so the screwed connections have to be retightened after letting off of pressure.

Is the line system tight throttle valves, pressure control valves, quantity valves etc. will be adjusted. The system can be checked for function at full working pressure and can be run/driven after this.

During running-in the filter inserts have to be cleaned within short intervals.

The first filter controls shall be done after 1 hour at the latest, after 12 hours and after 24 hours working time. After this the controls can be extended up to a maintenance time in rotation, which cannot be fixed generally, as this is dependent of the respective working conditions such as switching-on period of the system, external influences etc.. Furthermore the working temperature has to be observed during running-in. The working temperature should amount from 40 °C to 55 °C in general. In special cases there are higher temperatures possible, but here the responsible department in our works should be informed.

X. Inspection and Maintenance

During the working time the system has to be permanently observed and maintained. The inspection- and maintenance works are appropriately carried out according to an exactly worked out plan which determines the kind of inspections as well as their temporary interval/s. As these things are dependent on the respective working conditions we are in the position to propose a tabulated plan here in this place in which the individual measurements are not fixed temporally. But, coming up to the experience of the first working time – in which a permanent observation is necessary – we recommend to complete this process by completion of maintenance (intervals) periods.

Inspection	Maintenance
<i>Oil reservoir</i>	<i>Oil reservoir</i>
recheck oil level	clean fluid (rinse)
check cleanliness of liquid	oil change – if necessary
check oil temperature	refill missing liquid

<i>Filter- and Line Systems</i>	<i>Filter- and Line Systems</i>
control of filter inserts checking of lines for leakage inspection of lines for damaging inspection of lines for folds	cleaning of filter inserts exchange of inserts – if necessary retightening of screwed connections exchange of damaged lines exchange of damaged screwed connections
<i>Pumps and Control Devices</i>	<i>Pumps and Control Devices</i>
checking for external damage/s checking of tear condition checking of washers checking of pressure generation checking of adjustment pressures checking of the running and controlling behaviour	cleaning of individual parts checking of running surfaces exchange of wear and tear parts exchange of washers
<i>Miscellaneous Devices</i>	<i>Miscellaneous Devices</i>
check of all measuring devices check of electrical system check of fuel gas pressure checking of heat exchangers	

XI. Putting out of Action concerning longer Down-times and Resetting into Operation

Concerning a longer putting out of operation the oil reservoir has to be filled-up so that the oil mirror at the filler neck will reach the height of the reservoir (tank) cap, in order to protect the internal space against corrosion. Because of the same reason we recommend to have the system fully run through from time to time, in order to have all internal spaces of the hydraulic system moistened with hydraulic oil. Recommissioning has to be done and carried out like the first commissioning. Moreover the following points must be absolutely obeyed:

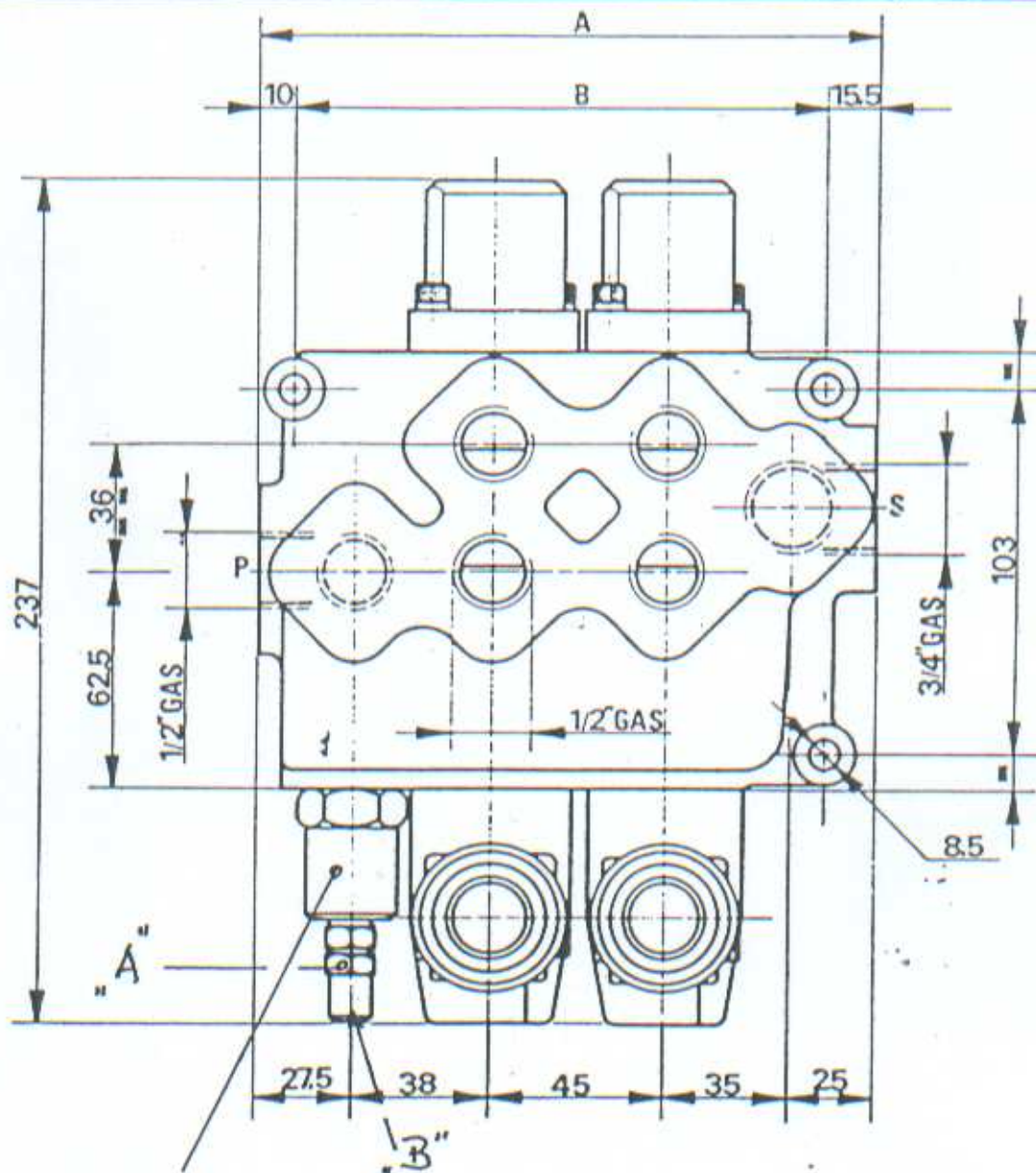
1. Control of pressure fluid concerning water contents, removal of contaminations. Exchange oil filling – if necessary.
2. Check oil level.
3. Check and clean sieves, filter/s and magnet separator/s and have them exchanged respectively.

For special Obeying

Concerning checking and overhauling of pumps and control devices the instructions within the attached service information of the individual devices have to be strictly obeyed. Please note that we cannot grant a guarantee concerning any improper handling / treatment.



Einstellung Druckbegrenzungsventil
Adjustment pressure limit device



Druckbegrenzungsventil / Pressure limit device

Druck einstellen :

1. Mit Sechskantmaulschlüssel 13 Mutter "A" lösen.
2. Mit Inbusschlüssel 6 Gewindestift "B" in Richtung des Ventiles einschrauben (bei laufendem Pumpenmotor)
3. Am Manometer Druck $p_{max} = 215$ bar ablesen / einstellen.
4. Sechskantmutter wieder festziehen.
5. Zylinder darf nicht über $p_{max} = 215$ bar beaufschlagt werden.

Adjustment of pressure :

1. Loose female screw "A" with hexagon spanner no.13.
2. Screw up threaded pin "B" with Inbus key no.6 into direction of the valve (pump engine on).
3. adjust pressure $p_{max.} = 215$ bar at the manometers.
4. Screw up hexagon nut again
5. Do not exceed cylinder pressure of $p_{max.} = 215$ bar.